

Reflectance Spectroscopy and Landsat Ground Truthing: Linking Material Spectra, NDVI, and Mixed-Pixel Effects

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Abstract

Reflectance spectroscopy connects material-scale optical properties to the multispectral measurements used in satellite remote sensing. This analysis integrates portable spectroradiometer measurements from 350 to 2500 nm with Landsat 8/9 Operational Land Imager (OLI) observations of Middletown, Connecticut. Field spectra distinguish healthy vegetation from pavement, soil, painted courts, and senescent plant material through the contrast between red and near-infrared reflectance. Using Landsat-compatible red and near-infrared wavelength windows, calculated NDVI values are highest for green vegetation and remain near zero for most nonvegetated surfaces. A comparison with Landsat shows good agreement over large, homogeneous grass cover, but strong disagreement over the small Dresser Diamond infield because a 30 m OLI pixel mixes dirt with surrounding turf. The same spatial-mixing effect appears in tennis-court, sidewalk, and Dresser Diamond spectra. Together, these results show that ground spectroscopy is most effective when spectral shape, sensor bandpass, spatial resolution, acquisition timing, and sub-pixel land-cover mixtures are interpreted jointly.

Introduction

Reflectance spectroscopy measures the fraction of incident electromagnetic energy reflected by a material as a function of wavelength. Because vegetation, soils, pavement, coatings, and other surface materials interact differently with visible, near-infrared (NIR), and shortwave-infrared (SWIR) radiation, their spectra can be used as diagnostic signatures. Portable spectroscopy provides a high-spectral-resolution view of individual targets, while orbital sensors such as Landsat observe the same landscape through a smaller number of broad spectral bands and much larger ground footprints. Comparing the two therefore provides both a material-identification test and a direct demonstration of scale effects in remote sensing.

The central comparison in this study is between point-scale spectra and Landsat 8/9 OLI measurements. The OLI red band is Band 4 and the NIR band is Band 5; for the densely sampled spectroradiometer data, the closest wavelength matches are averages over approximately 640-670 nm for red and 850-880 nm for NIR. These matched windows allow a consistent Normalized Difference Vegetation Index (NDVI) comparison while preserving the much finer spectral information available from the field instrument.

Methodology

Spectral Sampling and NDVI

Spectra were collected with an ASD FieldSpec FR portable spectroradiometer, which spans approximately 350-2500 nm. Multiple measurements of each target were averaged to reduce short-term noise and create a representative spectrum. The targets include asphalt, Dresser Diamond dirt and grass, green and

senescent vegetation, sidewalk, tennis-court surfaces, and track material. For comparisons with OLI, the high-resolution spectra were resampled to Landsat-compatible wavelength regions.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

NDVI is sensitive to the strong increase in healthy vegetation reflectance between red wavelengths, where chlorophyll absorption is important, and the NIR, where leaf internal structure produces high scattering. Values approaching 1 therefore indicate a large NIR-to-red contrast, whereas many mineral, pavement, and painted surfaces have similar red and NIR reflectance and consequently yield values near zero. Negative values can occur when red reflectance exceeds NIR reflectance.

Landsat Comparison and Spatial Scale

A recent Landsat 8/9 scene over Middletown was examined as true-color imagery and as an NDVI raster. Regions of interest were compared with the corresponding field targets. Landsat multispectral pixels are approximately 30 m on a side, so each measured pixel can combine multiple materials that would be separated by the much smaller field-spectrometer footprint. At a nominal orbital altitude of about 705 km, a 30 m ground footprint corresponds to an angular instantaneous field of view of roughly 43 microradians; the 15 m panchromatic channel corresponds to about 21 microradians. This scale difference is critical for interpreting agreement or disagreement between the two datasets.

Results

Field Spectral Library

The resampled spectral library separates vegetation from most built and bare-ground materials most clearly near the red-to-NIR transition. Green grass, green pine needles, and a green sweetgum leaf show low red reflectance followed by a strong NIR rise, while asphalt, concrete, court coatings, and track surfaces remain comparatively flat. Colored or senescent plant material retains some vegetation-like structure but shows a smaller NIR contrast than healthy green vegetation. Figure 1 summarizes this range of spectral behavior across the measured materials.

The NDVI calculations quantify the same pattern. Green pine needles have the highest measured value (0.835), followed by grass (0.809), green sweetgum leaf (0.736), and sugar gum fruit (0.696). Red maple leaves and dead pine needles are lower, consistent with reduced green-leaf spectral contrast. Most nonvegetated surfaces cluster close to zero, including sidewalk (0.030), track (0.058), red tennis court (0.059), and gray concrete court (-0.019). Dresser Diamond dirt is also low (0.142), well below the nearby grass signature.

Table 1: Representative red and NIR reflectance values and calculated NDVI.

Material	Red	NIR	NDVI
Asphalt 1	0.069	0.072	0.026
Dresser Diamond (Dirt)	0.117	0.155	0.142
Dresser Diamond (Grass)	0.065	0.617	0.809
Green American Sweetgum Leaf	0.077	0.506	0.736
Pine Needles (Dead)	0.299	0.496	0.247
Pine Needles (Green)	0.045	0.503	0.835
Sidewalk (Church St.)	0.058	0.061	0.030
Tennis Court (Gray Concrete)	0.144	0.139	-0.019
Track Surface	0.248	0.279	0.058

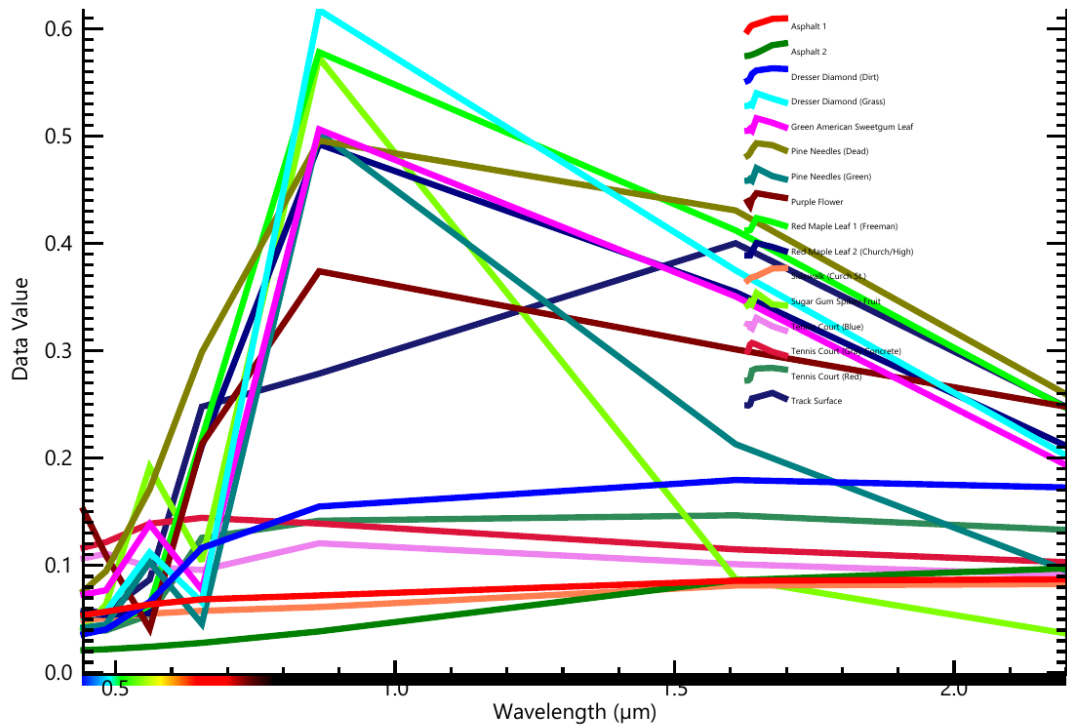


Fig. 1: Resampled reflectance spectra for the measured surface materials. Healthy green vegetation shows the strongest rise into the NIR, whereas most pavement, soil, and painted surfaces remain flatter and lower-contrast across the red-NIR transition.

Ground Truthing with Landsat NDVI

The Middletown true-color image provides the spatial context for the NDVI product, while the NDVI raster emphasizes the contrast between vegetated and nonvegetated surfaces across the same area. The mapped pattern is consistent with the field spectral library: vegetated land cover produces stronger positive NDVI than water, pavement, bare surfaces, and many developed areas. Figure 2 shows the two products at the same regional scale.

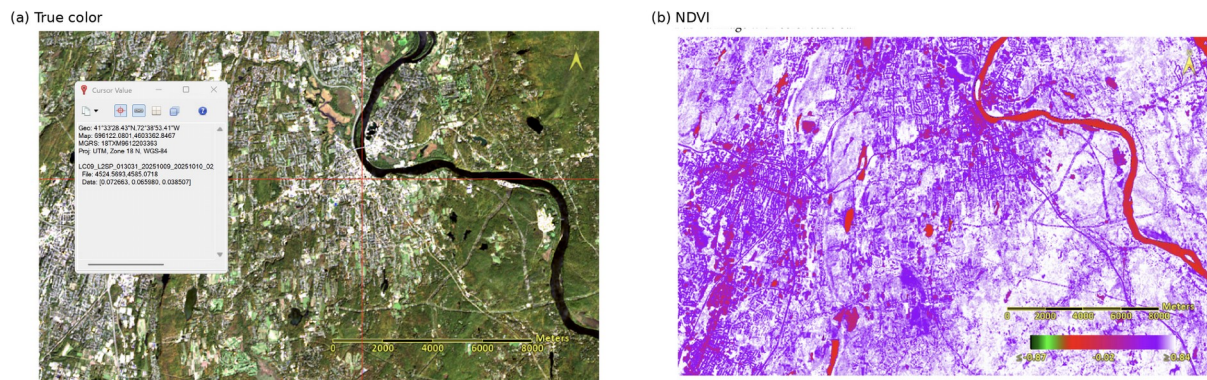


Fig. 2: Landsat true-color view of Middletown (left) and the corresponding NDVI image (right). The NDVI product reorganizes the scene by red-NIR contrast, making vegetation response easier to distinguish from many built, bare, and water surfaces.

At Andrus Field, the field vegetation NDVI (about 0.81) and Landsat ROI mean (about 0.86) agree closely, which is expected for a large, comparatively homogeneous grass surface. Dresser Diamond behaves differently: the dirt spectrum gives an NDVI of about 0.14, but the Landsat ROI mean is about 0.70. This is not a failure of NDVI; it is a scale mismatch. The 30 m Landsat pixels overlap the small dirt infield and the surrounding green turf, so the satellite measurement contains a strong vegetation contribution that the point-scale dirt measurement does not.

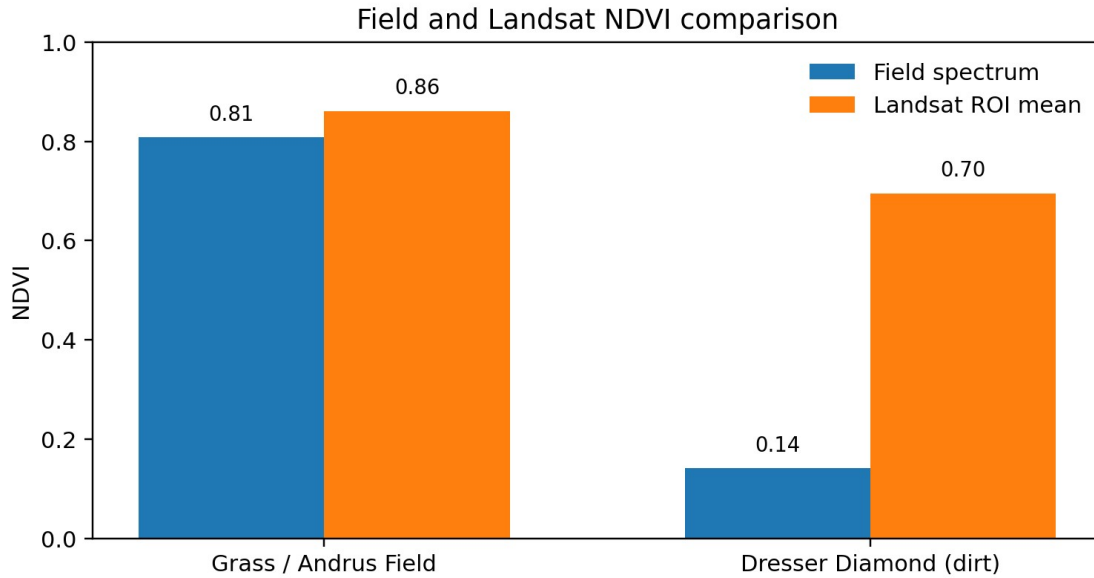


Fig. 3: Field and Landsat NDVI for grass/Andrus Field and Dresser Diamond dirt. Agreement is strong for the spatially extensive grass target, while the small dirt infield is strongly biased toward vegetation in the 30 m Landsat measurement because of sub-pixel mixing.

Spectral Mixing in Built Surfaces

The tennis-court comparison shows the same spatial-resolution effect in spectral rather than index form. The Landsat ROI mean broadly follows the blue-court spectral shape, but its reflectance magnitude differs from the point-scale blue surface, particularly from the red through SWIR. The Landsat pixel can include multiple court coatings, surrounding surfaces, small vegetation fractions, and illumination differences, so the satellite spectrum should be interpreted as an areal mixture rather than as a pure material spectrum.

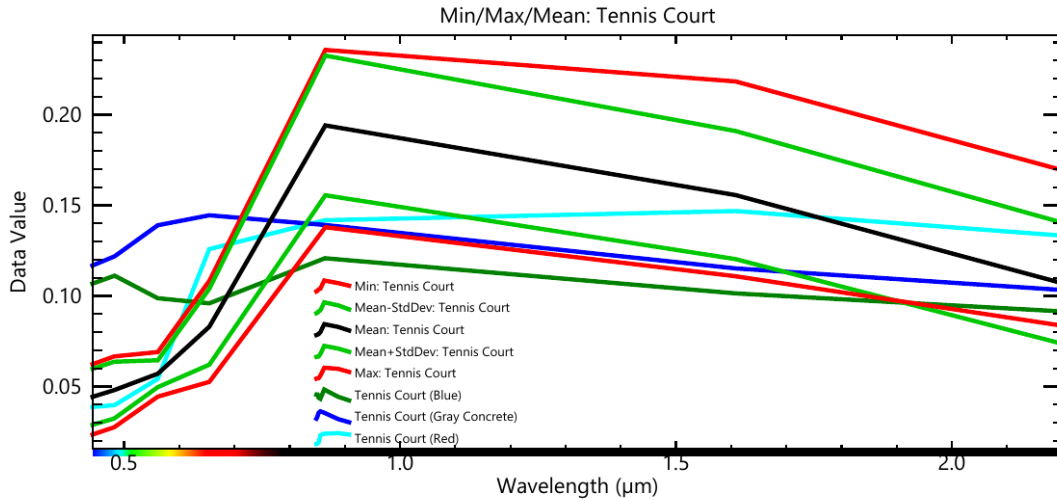


Fig. 4: Landsat ROI statistics for the tennis court compared with field spectra from blue, gray-concrete, and red court surfaces. The satellite mean preserves part of the court-scale spectral shape but differs in magnitude because each 30 m pixel integrates multiple surface components.

The Church Street sidewalk comparison is similarly pavement-like: both the field spectrum and Landsat ROI lack the pronounced red-to-NIR jump characteristic of healthy vegetation. The Landsat mean is nevertheless substantially brighter across much of the sampled range. The wide ROI envelope is consistent with spatial heterogeneity within and around the sidewalk, including adjacent curb or concrete surfaces and possible vegetation contributions.

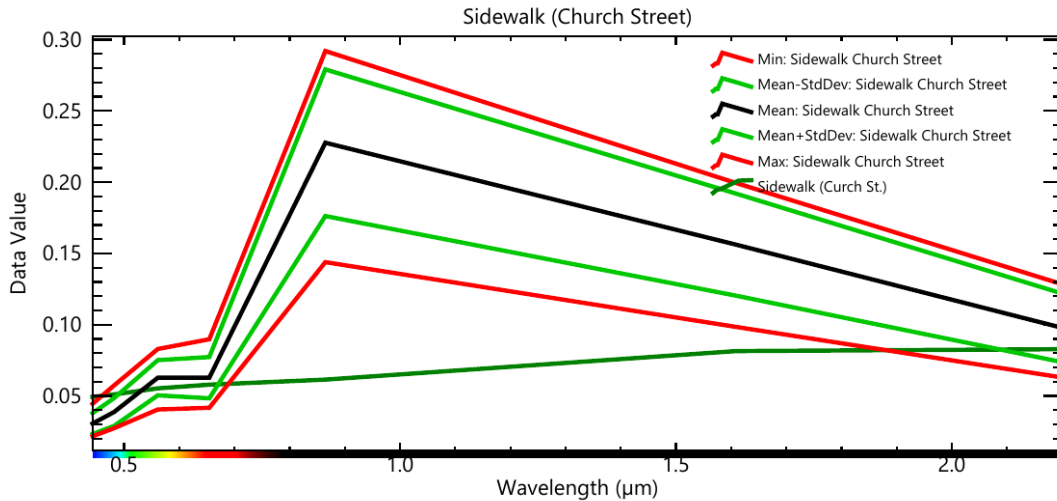


Fig. 5: Church Street sidewalk field spectrum compared with Landsat ROI statistics. The overall nonvegetated spectral form is retained, but the orbital measurement is brighter and more variable because the pixel footprint includes neighboring materials.

Dresser Diamond provides the clearest demonstration of what is contained in a Landsat pixel. The Landsat ROI has a pronounced red minimum and high NIR reflectance, with mean Band 4 reflectance near

0.0665 and mean Band 5 reflectance near 0.3708. That shape is much closer to the grass spectrum than to the dirt spectrum. Pixels centered more strongly on the infield trend toward the dirt response, while pixels overlapping more turf shift toward the vegetation endmember. Figure 6 therefore explains both the elevated Landsat NDVI and the difference between point and orbital spectra at this site.

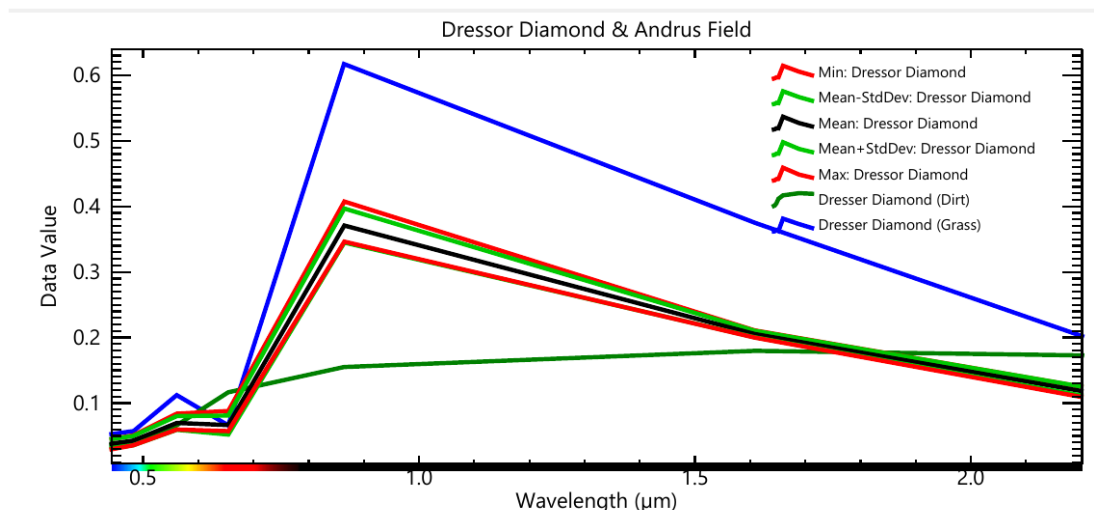


Fig. 6: Landsat Dressor Diamond ROI statistics compared with field grass and dirt spectra. The Landsat mean is substantially more vegetation-like than the pure dirt spectrum, demonstrating how the surrounding turf dominates a mixed 30 m pixel.

Discussion

Across the dataset, the most reliable correspondence between field spectroscopy and Landsat occurs when the target is spectrally distinctive and occupies a large fraction of the satellite pixel. Healthy grass satisfies both conditions: its low red reflectance and high NIR reflectance produce a strong vegetation signature, and Andrus Field is spatially extensive enough that many pixels are dominated by grass. Small or narrow targets such as the Dressor Diamond infield and sidewalk violate the second condition, so the satellite spectrum becomes an areal average of multiple materials.

The comparison also illustrates why spectral resolution and spatial resolution must be considered together. The ASD instrument provides dense measurements from 350 to 2500 nm and can isolate a single target, while OLI reduces that continuous spectrum to a limited set of bands over a much larger footprint. Resampling the field spectra into Landsat-compatible bands makes the spectral comparison physically meaningful, but it cannot remove the spatial mixture inherent to the orbital data. Differences in illumination, surface anisotropy, atmospheric path effects, and acquisition timing can further alter reflectance magnitude, especially for heterogeneous or directional surfaces.

Temporal matching is therefore important when using field spectra as ground truth. Vegetation phenology, moisture state, mowing or maintenance, shadows, and surface condition can change over days to seasons. A recent, cloud-free Landsat image minimizes these differences and makes disagreement easier to attribute to sensor scale and spatial mixing rather than to real change in the landscape.

Conclusion

The integrated field and Landsat analysis shows that reflectance spectroscopy is effective for distinguishing broad material classes and for interpreting vegetation indices, but only when sensor scale is treated explicitly. Healthy green vegetation exhibits the strongest red-to-NIR contrast and the highest NDVI values, while most pavement, court, track, and bare-soil surfaces remain near zero. Landsat reproduces the field vegetation response well over homogeneous grass, yet small targets can differ dramatically because a 30 m pixel records all illuminated materials within its footprint. The Dresser Diamond result is the clearest example: a dirt target with field NDVI near 0.14 appears near 0.70 in Landsat because surrounding turf dominates the mixed pixel. Tennis-court and sidewalk comparisons show the same principle in full spectral form. Ground truthing is therefore not a direct one-to-one transfer from a point spectrum to a satellite pixel; it is a scale-aware comparison that must account for bandpass, spatial resolution, target geometry, timing, and spectral mixture.

References

- U.S. Geological Survey (USGS). Landsat 8-9 Operational Land Imager (OLI) band designations and spatial-resolution specifications.
- Analytical Spectral Devices (ASD). FieldSpec FR portable spectroradiometer technical specifications (350-2500 nm spectral range).